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Micropolar effects on the effective elastic properties and elastic fracture toughness of planar lattices

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ABSTRACT

Effective elastic properties, and mode I elastic fracture toughness of three isotropic planar lattices – hexagonal, kagome, and triangular – are studied from a micropolar continuum perspective. The hexagonal lattice is bending dominated whereas the kagome and the triangular are stretching dominated for any prescribed macroscopic in-plane loading. Discrete asymptotic homogenization is applied to obtain analytical expressions for the topology governed effective micropolar elastic properties of each lattice and their scaling with relative density: the hexagonal lattice is found to possess the largest micropolar internal length parameter at any given relative density. The homogenized micropolar continuum is then discretized by a four-node finite element model to compute its Cauchy and couple-stress intensities for a central crack under mode I loading. Micropolar scaling exponents for mode I fracture toughness with relative density agree well with the estimates from the literature based on fully discrete beam network models. Hexagonal lattice exhibits an order of magnitude higher couple stress intensity factor when compared to the triangular and the kagome lattices of identical relative density, signifying the importance of micropolar effects in bending dominated architectures.

1. Introduction

Architected metamaterials exploit topology governed properties as a design principle to access gaps in material property space (Ashby and Brachet, 2003; Fleck et al., 2010; Phani and Hussein, 2017; Schaedler and Carter, 2016; Kadic et al., 2019) that otherwise are unattainable using traditional materials. Architected materials are realized by advances in manufacturing techniques. They find applications in several areas, including in multifunctional structures (Greer and Deshpande, 2019; Mueller et al., 2018), as biomedical devices (Douglas et al., 2014), and in energy applications (Tappan et al., 2010). Linking the geometry to macroscopic properties is important to assess their potential for applications. Discrete beam network calculations and analytical homogenized continuum methods have been used to identify the functional dependence of effective elastic properties and mode I fracture toughness on lattice porosity. However, discrepancies are noted between the continuum model predictions for fracture toughness and the discrete beam network calculations (Fleck and Qiu, 2007; Chen et al., 1998). This study resolves these by correctly accounting for bending and stretching deformation in a micropolar continuum framework. Analytical expressions are given for porosity dependent scaling of effective properties, Cauchy and moment stress intensities, and the micropolar internal length parameter considered as an intrinsic property of each topology.

Bounds on macroscopic effective elastic material properties of lattices can be obtained by modeling them as multi-phase porous composite solids (Hashin and Shtrikman, 1963; Milton, 2002; Yeong and Torquato, 1998), or by treating them as a spatially periodic network of struts and then invoking the principles of structural mechanics of beam frameworks (Gibson and Ashby, 1997; Gibson, 2005). Both approaches are equivalent in the low relative density (high porosity) limit. Relative density, denoted by $\bar{\rho}$, is defined here as a ratio of the densities of the porous solid and its parent material; it takes values in the closed interval $0 \leq \bar{\rho} \leq 1$. The notion of bending and stretching dominance has prevailed in the classification of architected materials (Fleck et al., 2010). It has been recognized that stretching dominated geometries possess high specific stiffness. Maxwell's rule for simply stiff networks (Pellegrino and Calladine, 1986; Guest and Hutchinson, 2003) is invoked to account for the cell geometry. We will explicitly show how these ideas carry into a micropolar continuum description of lattices by considering three isotropic lattices: the

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hexagonal lattice (bending dominated), and the kagome and the triangular lattices (stretching dominated). Our choice necessarily restricts to isotropic micropolar continua. Even though higher order continuum description of a hexagonal lattice has been attempted earlier for effective elastic properties (Pradel, 1998; Wang and Stronge, 1999), the scaling of effective elastic fracture toughness on relative density remains to be addressed. It is of interest whether the distinction between bending and stretching dominated lattices governing the scaling relation between Cauchy moduli and relative density (Deshpande et al., 2001; Hutchinson and Fleck, 2006) carries over to the relative density dependence of micropolar constitutive parameters. We address this by deriving explicit relations through an extended discrete asymptotic homogenization (Sanchez-Hubert and Sanchez-Palencia, 1992; Mourad, 2003; Warren and Byskov, 2008; Dos-Reis, 2010; Pradel, 1998).

Intuitively, one expects the toughness scaling to transition from a porous solid ($\overline{\rho} < 1$) to that of the parent solid material (at $\overline{\rho} = 1$). While bounds on properties such as moduli can be set, fracture toughness remains unbounded due to length scale dependent toughening mechanisms (Alonso and Fleck, 2009). Within a linear elastic fracture mechanics setting, predictions from gradient continuum theories (Chen et al., 1998) can give incorrect scaling for fracture toughness (Fleck and Qiu, 2007) with relative density, if non-affine deformation due to strut bending is not taken into account. The present framework offers a unified description of toughness dependence on relative density. We show that micropolar continuum model not only gives scaling results consistent with structural mechanics at low relative density (Fleck and Qiu, 2007) but also the scaling results extend to higher ranges of relative densities, where structural beam mechanics assumptions break down. The beam approximation of the lattice cell walls is no longer tenable at higher relative densities, since the cell wall deformations are that of a three dimensional solid. Discrete asymptotic homogenization method (Sanchez-Hubert and Sanchez-Palencia, 1992; Mourad, 2003; Warren and Byskov, 2008; Dos-Reis, 2010; Pradel, 1998) is adapted to formulate a higher order continuum model for such lattice materials. This approach yields explicit relations between the effective micropolar constitutive parameters and the relative density for each topology. Thereby, the micropolar bending effects will be incorporated into the fracture criterion by introducing a micropolar stress intensity factor into the energy release rate. These relations are used subsequently to compute the relative density dependence of fracture toughness using a finite element discretization of a square domain with a central crack.

In Section 2, micropolar stiffness matrices are presented for the three isotropic centro-symmetric lattices. We then apply a discrete asymptotic homogenization method in Section 3 to obtain explicit relations for all effective properties for the three lattice topologies by formulating their stiffness matrices. The method will be illustrated for a hexagonal lattice. Relative density dependent effective elastic properties of a micropolar continuum are obtained which are used in the subsequent calculations for fracture toughness. The explicit relations for effective micropolar parameters are used in Section 4 to formulate a finite element model of the homogenized micropolar lattice. The finite element model of the higher order continuum is then used to calculate mode I fracture toughness for a central crack under remote tension loading. Calculated values for mode I toughness obtained from the discretized micropolar continuum are compared with fully discrete beam lattice simulations from the literature. The relative contribution of couple stress intensities will be assessed for each lattice topology. Section 5 summarizes the main findings of this study.

2. Micropolar stiffness matrices

Micropolar constitutive models for periodic lattices are motivated by the existence of non-affine bending deformation mechanisms due to the very low bending modulus of the slender structural lattice elements. The individual strut deformations do not follow the macroscopically imposed kinematics of deformation: two initially parallel struts will not remain parallel after deformation. The consideration of microrotation as an additional continuum degree of freedom entails the existence of bending moments due to the unbalanced internal moments caused by the rotation and bending of the structural elements at the microlevel. Microstructural effects have been observed to be important when sample dimensions are comparable with the unit cell size (Park and Lakes, 1986), leading to so called "size effects" (Onck et al., 2001; Tekoğlu and Onck, 2008; Eremeyev and Pietraszkiewicz, 2015). In classical continuum mechanics (Truesdell and Noll, 1992), all higher order displacement gradients are neglected and hence size effects are not admissible in a Cauchy continuum (Truesdell and Noll, 1992). First gradient Cauchy theory needs to be enriched in situations when the loading wavelength or the wavelength of the deformation field approaches the typical microstructure size. In such situations where the strict length scale separation assumption fails, Cauchy continuum is enhanced by incorporating additional intrinsic parameters and internal length scales associated with the microstructure.

For a plane problem with z as the out of plane axis, the stress tensor has four independent components σ_{xx}, σ_{yy}, σ_{xy}, σ_{yx} ($\sigma_{yx} \neq \sigma_{xy}$) and the couple-stress has two independent components m_{xz} and m_{yz}. Similarly, classical strain components e_{xx}, e_{yy}, e_{xy}, e_{yx}, and microcurvature components κ_{xz} and κ_{yz} are the work-conjugate strain variables. Denoting the displacement field by $\mathbf{u}$ and microrotation field by $\boldsymbol{\phi}$, the deformation and curvature tensors in component form follow as Eringen (1966)

$$e_{ij} = u_{i,j} + \varepsilon_{ijk}\phi_k; \;\; \kappa_{ij} = \phi_{i,j}, \tag{1}$$

where ε_{ijk} is the Levi-Civita permutation tensor and the comma in the subscript denotes partial derivative with respect to the symbol following it. The curvature-deformation compatibility equations follow as:

$$\kappa_{iz} = \varepsilon_{ijz}\left(e_{ji,i} - e_{ii,j}\right); \;\; \varepsilon_{jiz}\kappa_{iz,j} = 0 \tag{2}$$

Equilibrium equations resulting from momentum balance for the translation and rotation are given by

$$\sigma_{ij,j} + F_i = 0 \tag{3}$$

$$m_{ik,i} + \varepsilon_{ijk}\sigma_{ij} + Q_k = 0 \tag{4}$$

where F_i are the body forces, while Q_k is the body couple. Eq. (4) shows that the antisymmetric part of the stress tensor σ_{ij} is determined by the divergence of the couple-stress tensor m_{ij}. Constitutive equations of a linear anisotropic micropolar elastic medium are of the form

$$\sigma_{ij} = A_{ijkl}e_{kl} + B_{ijkl}\kappa_{kl} \tag{5}$$

$$m_{ij} = C_{ijkl}e_{kl} + D_{ijkl}\kappa_{kl}, \tag{6}$$

where the fourth order tensors A_{ijkl}, B_{ijkl}, C_{ijkl} and D_{ijkl} are called micropolar stiffness tensors; B_{ijkl} and C_{ijkl} are coupling pseudo tensors that vanish for periodic structures with central symmetry (Trovalusci and Masiani, 1999; Kumar and McDowell, 2004, 2009) and they are related by $\mathbf{C} = \mathbf{B}^T$. A_{ijkl} and D_{ijkl} satisfy the index symmetry properties $A_{ijkl} = A_{klij}$, $D_{ijkl} = D_{klij}$. For a linear isotropic micropolar material, the constitutive equations reduce to

$$\sigma_{ij} = \lambda e_{ll}\delta_{ij} + (\mu^* + \mu_c)e_{ij} + \mu^* e_{ji} \tag{7}$$

$$m_{ij} = \alpha\phi_{l,l}\delta_{ij} + \beta\phi_{i,j} + \gamma\phi_{j,i} \tag{8}$$

where μ_c, α, β, γ are new micropolar constants in addition to the classical Lamé constants λ and μ with $\mu^* = \mu - (\mu_c/2)$. These effective micropolar constants depend not only on the network geometry of the unit cell but also on its relative density. This relationship is succinctly expressed in terms of the relative density parameter. Thus, the functional dependence of μ_c, α, β, γ on $\overline{\rho}$ needs to be established first.

The constitutive equations for the equivalent 2D isotropic micropolar continuum can be written in matrix format as

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \\ \sigma_{yx} \\ m_{xz} \\ m_{yz} \end{bmatrix} = \begin{bmatrix} \boldsymbol{A}_e & 0 \\ 0 & \boldsymbol{D}_e \end{bmatrix} \begin{bmatrix} e_{xx} \\ e_{yy} \\ e_{xy} \\ e_{yx} \\ \kappa_{xz} \\ \kappa_{yz} \end{bmatrix} \tag{9}$$

where, $\boldsymbol{A}_e$ is the 4×4 Cauchy stiffness matrix and $\boldsymbol{D}_e$ is the 2×2 stiffness matrix that relates the couple stress to the curvature.

For a plane problem and a central symmetric lattice, the stiffness matrix is defined according to the above constitutive law by four independent constants : λ, μ^*, μ_c and γ (Fatemi et al., 2002; Cowin and Pennington, 1970). The lattices considered here are similar to continuous material with plane strain and do not have a Poisson's ratio in the thickness direction (normal to the plane) and the stiffness matrix of a centrally symmetric material is of the form (Trovalusci and Masiani, 1999; Yang et al., 2003)

$$\begin{bmatrix} \frac{(2\mu^*+\mu_c)(2\lambda+2\mu^*+\mu_c)}{\lambda+2\mu^*+\mu_c} & \frac{\lambda(2\mu^*+\mu_c)}{\lambda+2\mu^*+\mu_c} & 0 & 0 & 0 & 0 \\ \frac{\lambda(2\mu^*+\mu_c)}{\lambda+2\mu^*+\mu_c} & \frac{(2\mu^*+\mu_c)(2\lambda+2\mu^*+\mu_c)}{\lambda+2\mu^*+\mu_c} & 0 & 0 & 0 & 0 \\ 0 & 0 & \mu^*+\mu_c & \mu^* & 0 & 0 \\ 0 & 0 & \mu^* & \mu^*+\mu_c & 0 & 0 \\ 0 & 0 & 0 & 0 & \gamma & 0 \\ 0 & 0 & 0 & 0 & 0 & \gamma \end{bmatrix} \tag{10}$$

From the compliance matrix, the micropolar Young's modulus, the homogenized shear modulus, the Poisson's ratio and the internal characteristic length express successively as follows using a star superscript notation for the effective moduli

$$E^* = \frac{(2\mu^*+\mu_c)(3\lambda+2\mu^*+\mu_c)}{2\lambda+2\mu^*+\mu_c} \tag{11}$$

$$G^* = \mu^* + \frac{\mu_c}{2} \tag{12}$$

$$v^* = \frac{\lambda}{2\lambda+2\mu^*+\mu_c} \tag{13}$$

$$l_c = \sqrt{\frac{\gamma}{2(2\mu^*+\mu_c)}} \tag{14}$$

As an aside, a special case of the micropolar theory, in which the microrotations are not independent kinematic variables but are defined in terms of the displacement gradients has been developed by Koiter under the name couple stress theory (Koiter, 1964). A continuous transition from couple-stress to micropolar theory is analyzed by Cowin and Pennington (1970) considering a coupling number N, where $N = 0$ corresponds to the classical theory, $N = 1$ corresponds to the couple stress theory, and $0 < N < 1$ corresponds to the micropolar theory, with

$$N = \sqrt{\frac{\mu_c}{2(\mu^*+\mu_c)}} \tag{15}$$

The necessity to maintain a positive internal energy for stability further imposes the following conditions on the micropolar constants

$$3\lambda+2\mu^*+\mu_c \geq 0,\quad 2\mu^*+\mu_c \geq 0,\quad 3\alpha+\beta+\gamma \geq 0,\quad -\gamma \leq \beta \leq \gamma,\quad \mu_c \geq 0 \tag{16}$$

3. Effective micropolar constants of the three lattices

The homogenization of elastic periodic discrete materials has been investigated in earlier contributions (Sab, 1996; Pradel and Sab, 1998; Pradel, 1998). Asymptotic expansion methods and other related homogenization techniques were used to show that the overall elastic properties of a beam lattice can be derived by solving a discrete unit cell problem involving a finite number of degrees of freedom. In this contribution, the discrete asymptotic homogenization (DAH) method proposed in Tollenaere and Caillerie (1998), Caillerie et al. (2006), and in Reis and Ganghoffer (2012) is exploited for the construction of micropolar continua for elastic lattice materials endowed with translational and rotational degrees of freedom. The proposed discrete homogenization method treating the lattice using beam mechanics can handle the homogenization of repetitive unit cells including internal nodes; it is based on the asymptotic development of the kinematic and static variables under consideration, to an arbitrary order *a priori*. The method relies on a description of the connectivity of the lattices within the repeated reference lattice unit cell and on the mechanical equilibrium written at each node. The discrete asymptotic homogenization method is exposed in a synthetic manner in Appendix A.

We now focus on the homogenization of three different lattices: hexagonal, triangular and kagome shown in Fig. 1(a), (b), and (c), respectively. The crack of length $2a$ is inserted in the homogenized continuum for toughness calculations to follow. The aim is to replace the lattices by an equivalent micropolar continuum, shown in Fig. 1(d). We extend the discrete asymptotic homogenization method as a mathematical tool to homogenize the lattices in order to identify the micropolar constants of the equivalent generalized continuum.

3.1. Extension of the DAH method

Any infinite planar lattice can be obtained by repeating a reference unit cell by tessellating the plane along a set of basis vectors. Hence, we consider a family of lattices parameterized by a small scale parameter $\epsilon = \frac{l}{L}$ illustrated in Fig. 2; it is defined as the ratio of a characteristic strut length (l) of the reference unit cell to a characteristic lattice length (L). In the continuum limit, ϵ approaches 0 and the number of cells within the fixed reference domain increases continuously. Accordingly, the topology of any lattice is completely described by the identification of its reference unit cell, the parameterization of the underlying nodes and beams, and the two basis vectors (in 2D) providing the tessellation of the plane by the unit cell. Each beam is characterized by three geometrical parameters, its length l, width t and thickness w, shown in Fig. 2.

In order to formulate effective micropolar continuum's properties the discrete homogenization method is based on the asymptotic development of two kinematic parameters, the displacement $\mathbf{u}^n$ and the rotation at lattice nodes ϕ^n. The two geometrical parameters t (in-plane strut thickness) and w (thickness of strut in the out-of-plane direction) in Fig. 2 are considered different and decoupled ($w \neq t$), and in order to simplify somewhat the formulation without compromising the technical developments, one imposes a unit beam thickness $w = 1$. The moments depend on both tensile and flexural moduli k_l and k_f, defined as follows:

$$k_l = \frac{E_s S}{l} \quad \text{and} \quad k_f = \frac{12 E_s I}{l^3} \tag{17}$$

where E_s represents the elastic modulus of parent solid material, I the second moment of area of cross section of beam, and S the area of cross section of the beam. In Fig. 2 we observe that two length scale ratios exist in a lattice, the small classical scale parameter of asymptotic expansion $\epsilon = \frac{l}{L}$, and the small beam slenderness ratio $\eta = \frac{t}{l}$ which is imposed by the choice of a Timoshenko beam theory to account for shear deformations and connect with micropolar theories. In our case, the thickness is constant and independent of ϵ, then the tensile and flexural stiffnesses are expressed versus the slenderness ratio as

$$k_l = \frac{E_s t}{l} = E_s \eta \quad \text{and} \quad k_f = \frac{E_s t^3}{l^3} = E_s \eta^3 \tag{18}$$

The weak form of equilibrium of the lattice is written, with the asymptotic expansions of the kinematic and static variables inserted. The effective micropolar continuum is obtained by replacing discrete sums over lattice nodes and unit cells by Riemann integrals highlighting the stress and couple stress tensors (Reis and Ganghoffer, 2012). These static variables are related to the (non-symmetric) strain tensor and curvature tensor (the gradient of the microrotation), thereby highlighting the derived homogenized lattice constitutive law.

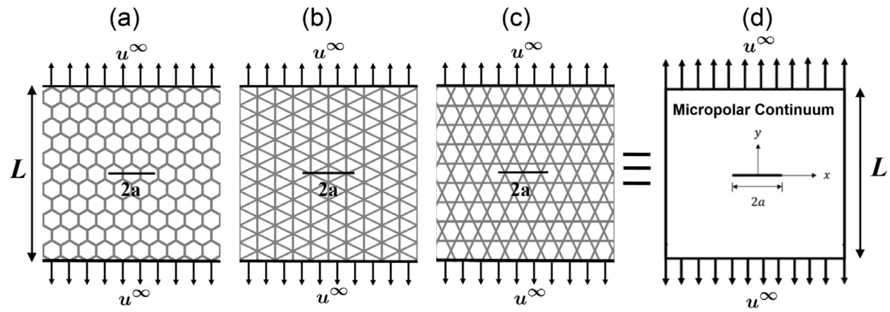

Fig. 1. Isotropic planar lattices considered in this study: (a) Hexagonal, (b) Triangular and (c) Kagome with a central crack subjected to mode I loading. All lattices are homogenized as a micropolar continuum shown in (d) whose effective properties change with the lattice geometry and relative density.

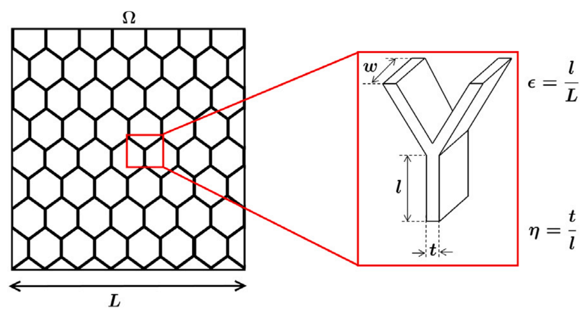

Fig. 2. Geometric parameters characterizing the different levels of the lattice microstructure.

Table 1
Micropolar constants of different lattices. The approximations for low relative density limits are compared with Fleck et al. (2010), Phani and Hussein (2017) and Gibson and Ashby (1997). Relative density, denoted by $\bar{\rho}$, is defined here as a ratio of the densities of the porous solid and its parent material; it takes values in the closed interval $0 \leq \bar{\rho} \leq 1$.

Parameter	Hexagonal	Triangular	Kagome
Relative density $\bar{\rho}$	$\frac{2}{\sqrt{3}}\frac{t}{l}$	$2\sqrt{3}\frac{t}{l}$	$\sqrt{3}\frac{t}{l}$
Young's modulus $\frac{E^*}{E_s}$	$\frac{3\bar{\rho}^3(8-6\bar{\rho}^2)}{(4+3\bar{\rho}^2)(4+9\bar{\rho}^2)} \approx \frac{3\bar{\rho}^3}{2}$	$\bar{\rho}\left(\frac{12+\bar{\rho}^2}{36+\bar{\rho}^2}\right) \approx \frac{\bar{\rho}}{3}$	$\bar{\rho}\left(\frac{6+\bar{\rho}^2}{18+\bar{\rho}^2}\right) \approx \frac{\bar{\rho}}{3}$
Shear modulus $\frac{G^*}{E_s}$	$\frac{3}{2}\frac{\bar{\rho}^2}{(4+3\bar{\rho}^2)} \approx \frac{3\bar{\rho}^2}{8}$	$\frac{\bar{\rho}}{96}\left(12+\bar{\rho}^2\right) \approx \frac{\bar{\rho}}{8}$	$\frac{\bar{\rho}}{48}\left(6+\bar{\rho}^2\right) \approx \frac{\bar{\rho}}{8}$
Poisson's ratio ν^*	$\frac{4-3\bar{\rho}^2}{4+9\bar{\rho}^2} \approx 1$	$\frac{12-\bar{\rho}^2}{36+\bar{\rho}^2} \approx \frac{1}{3}$	$\frac{6-\bar{\rho}^2}{18+\bar{\rho}^2} \approx \frac{1}{3}$
Internal length l_c/l	$\sqrt{\frac{\bar{\rho}(4+3\bar{\rho}^2)}{192}} \approx \sqrt{\frac{\bar{\rho}}{48}}$	$\frac{\bar{\rho}}{\sqrt{12(12+\bar{\rho}^2)}} \approx \frac{\bar{\rho}}{12}$	$\frac{\bar{\rho}}{\sqrt{6(6+\bar{\rho}^2)}} \approx \frac{\bar{\rho}}{6}$

3.2. Evaluation of the micropolar material constants

Based on the discrete asymptotic homogenization method (Reis and Ganghoffer, 2012), the effective stiffness matrix for the hexagonal, triangular and kagome lattices is evaluated in the B as a function of relative density. From these matrices, it is possible to derive the expression of the homogenized moduli as a function of the geometrical and lattice parameters. Effective Young's modulus, Shear modulus, Poisson's ratio and characteristic length are summarized in Table 1. We can observe that for slender beams ($l \gg t$, or $\bar{\rho} \ll 1$), the expression of the moduli for the hexagonal lattice becomes

$$E^* = \frac{3}{2}E_s\bar{\rho}^3, \quad G^* = \frac{3}{8}E_s\bar{\rho}^2, \quad \nu^* = 1 \tag{19}$$

which coincide with the expression given by various authors (Gibson and Ashby, 1997; Gibson, 2005). In order to compare the mechanical properties of the micropolar continuum for different type of lattices, we plot in Fig. 3 their evolution with the relative density. The predicted micropolar rotation field is in good agreement with the discrete beam rotation computed over the fully resolved lattice, as shown in Fig. C.10 of Appendix C.

The mechanical properties and internal bending lengths are computed beyond the low density regime considered in Gibson and Ashby (1997) and Gibson (2005). The triangular and kagome lattices show similar moduli with increasing relative density; however, their micropolar length parameter l_c is an order of magnitude lower compared to a hexagonal lattice. We also observe that the bending dominated hexagonal lattice is compliant compared to the stretching dominated triangular and kagome lattices. These results suggest that the bending dominated hexagonal lattice has a significantly higher micropolar length scale as the plot in Fig. 3 reveals.

4. Fracture toughness of homogenized micropolar lattices

Several authors have computed the fracture toughness of lattices. Maiti et al. (1984) assumed that the mode I fracture toughness of an elastic-brittle hexagonal lattice is dictated by the bending fracture of the cell beams immediately in front of the crack tip. They assumed that brittle fracture occurs when the maximum of the local tensile stress across the network reaches the tensile strength of the beam wall σ_f. The critical stress-intensity factor K_{IC} varies according to the following relationship

$$\frac{K_{IC}}{\sigma_f\sqrt{a}} = 0.4\bar{\rho}^2 \tag{20}$$

The same problem has subsequently been treated by Chen et al. (1998). These authors replaced the hexagonal lattice by a micropolar continuum and computed the asymptotic crack fields based on the affine deformation hypothesis and obtained a different relationship for fracture toughness

$$\frac{K_{IC}}{\sigma_f\sqrt{a}} = 1.56\bar{\rho}. \tag{21}$$

The above gives a higher toughness than Maiti's prediction (20) and experiments (Gibson and Ashby, 1997). Later, Fleck and Qiu (2007) calculated fracture toughness scaling with $\bar{\rho}$ and obtained the following power law

$$\frac{K_{IC}}{\sigma_f\sqrt{a}} = D\bar{\rho}^d. \tag{22}$$

They calculated the scaling exponent and the pre-factor in Fleck and Qiu (2007) using discrete beam network simulations. These results serve as a validation for our micropolar continuum model which is valid even at higher relative densities where beam network assumptions fail.

Consider now the elastic fracture toughness of the three lattices under a mode I loading condition, as sketched in Fig. 1. Recalling that gradient theories that do not account for non-affine deformations (Chen et al., 1998) have been shown to give incorrect scaling (Fleck and Qiu, 2007), it is of interest to investigate whether or not our micropolar continuum can reproduce existing scaling laws which have also been verified experimentally in the low relative density limit. The effective properties of the micropolar continuum obtained using discrete

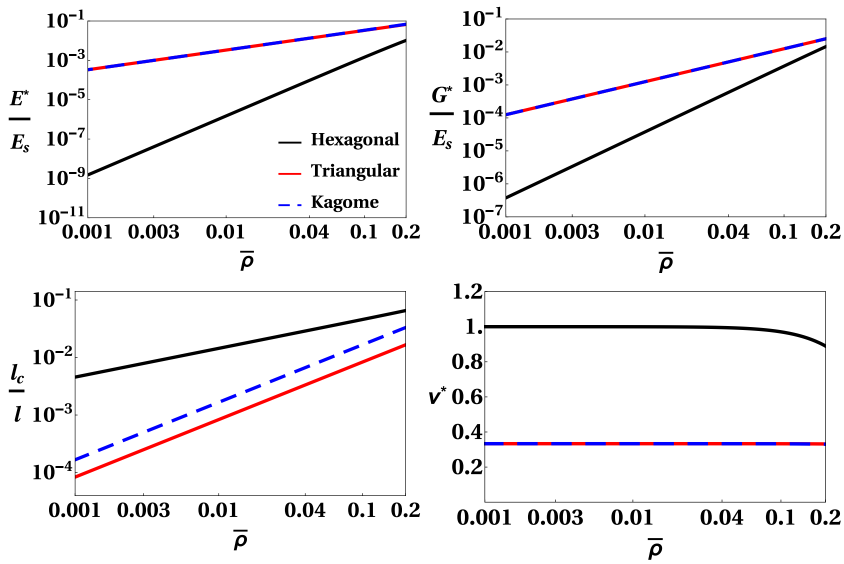

Fig. 3. Evolution of micropolar parameters versus relative density.

asymptotic homogenization, summarized in Table 1, are now used. The appropriate asymptotic displacement field expressions from Sternberg and Muki (1967) for the plane-strain couple stress problem using a polar coordinate system (r, θ) for a central crack of length $2a$ under a transverse field of uniform uni-axial tension, with a far field mode I loading σ^{∞} are,

$$\sigma_{xx}(r,\theta) = (2\nu-1)\frac{K_I}{\sqrt{2r}}\left[\cos\frac{\theta}{2} - \frac{1}{2}\sin\theta\sin\frac{3\theta}{2}\right] + O(r^0) \tag{23}$$

$$\sigma_{yy}(r,\theta) = \frac{K_I}{\sqrt{2r}}\left[(3-2\nu)\cos\frac{\theta}{2} - \frac{1}{2}(1-2\nu)\sin\theta\sin\frac{3\theta}{2}\right] + O(r^0) \tag{24}$$

$$\sigma_{xy}(r,\theta) = \frac{K_I}{\sqrt{2r}}\left[(4\nu-4)\sin\frac{\theta}{2} - \frac{1}{2}(1-2\nu)\sin\theta\cos\frac{3\theta}{2}\right] + O(r^0) \tag{25}$$

$$\sigma_{yx}(r,\theta) = (2\nu-1)\frac{K_I}{\sqrt{2r}}\left[\frac{1}{2}\sin\theta\cos\frac{3\theta}{2}\right] + O(r^0) \tag{26}$$

$$m_{xz}(r,\theta) = -\frac{L_I}{\sqrt{2r}}\left[\frac{a}{2}\sin\frac{\theta}{2}\right] + O(r^0) \tag{27}$$

$$m_{yz}(r,\theta) = \frac{L_I}{\sqrt{2r}}\left[\frac{a}{2}\cos\frac{\theta}{2}\right] + O(r^0) \tag{28}$$

where σ_{ij} are stresses, m_{ij} are couple-stresses, and ν is the Poisson's ratio. The stress and couple-stress intensity factors therein can be expressed as

$$K_I = f(\lambda,\nu)\sigma^{\infty}\sqrt{a}, \quad L_I = g(\lambda,\nu)\sigma^{\infty}\sqrt{a} \tag{29}$$

where $\lambda = l_c/a$, l_c is a characteristic length of the micropolar material and $2a$ is the crack length.

The asymptotic expressions of the displacement and rotation fields and the strain energy release rate ($\mathcal{G}$) are found by Sih and Liebowitz (1969) for a constrained micropolar medium ($N = 1$) in polar coordinates (Fig. 4) as

$$u(r,\theta) = \frac{K_I\sqrt{2r}}{8\mu^*}\left[(3-6\nu)\cos\frac{\theta}{2} + (6\nu-7)\cos\frac{3\theta}{2}\right] + O(r) \tag{30}$$

$$v(r,\theta) = \frac{K_I\sqrt{2r}}{8\mu^*}\left[(2\nu-1)\sin\frac{\theta}{2} + (7-6\nu)\sin\frac{3\theta}{2}\right] + O(r) \tag{31}$$

$$\phi(r,\theta) = \frac{L_I\sqrt{2r}}{8\mu^* l_c^2}\left[a\sin\frac{\theta}{2}\right] + O(r) \tag{32}$$

$$\mathcal{G} = \frac{\pi}{2\mu^*}\left[(1-\nu)(3-2\nu)K_I^2 + \frac{a^2}{16 l_c^2}L_I^2\right] \tag{33}$$

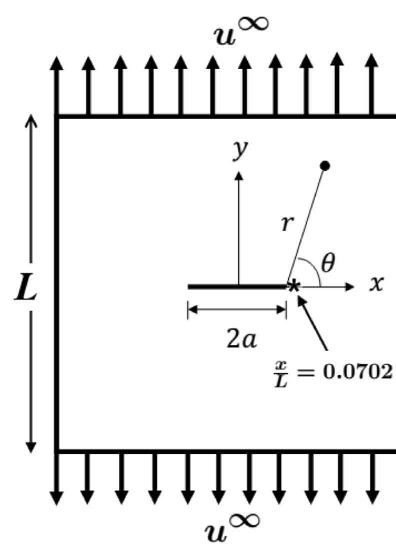

Fig. 4. Polar coordinate system used to calculate the displacement and stress field of the equivalent micropolar continuum, with the origin located at the center of the crack. The stress is evaluated slightly ahead of the crack tip at the point marked by a star, $*$, located at $\frac{x}{L} = 0.0702$. The crack size is $\frac{2a}{L} = 0.14$ so that the crack tip is located at $\frac{x}{L} = 0.07$. Vertical displacements are applied on the top and bottom edges of the specimen and the sides are free.

Similar expressions exist for mode II and mixed-mode conditions as well for $N \neq 1$ with different angular dependence functions, see for example Diegele et al. (2004) and Yu et al. (2014). We use the above displacement expressions to determine critical stress intensity under more I loading. We do this for each lattice with a central crack of length $2a$ for tensile loading conditions, as sketched in Fig. 1.

Objectives for our finite element calculations based on a homogenized micropolar continuum are to:

(a) evaluate the fracture toughness for each topology with varying relative density by using K-field displacements according to (30) and (31) and calculate K_{IC} associated with maximum tensile stress at a point slightly ahead of the crack tip exceeding the tensile strength of the material when the specimen is subjected to vertical displacements on the bottom and top edges and the sides are free;

(b) validate numerical results for toughness in the low relative density regime with the existing scaling laws in the literature, which are based on beam theories (Gibson and Ashby, 1997; Fleck and Qiu, 2007; Maiti et al., 1984);
(c) assess the influence of micropolar effects on fracture toughness, specifically in terms of the two parameters: the micropolar material length l_c and the couple-stress intensity L_I.

4.1. Finite element calculation of stress intensity factor

An effective micropolar continuum model is used as a substitute for each lattice based on the effective properties given in Table 1. We discretize the micropolar continuum by a four-node quadrilateral element by following the procedure outlined in Yang et al. (2003), and implemented in a finite element package (ABAQUS). Shear locking is a well-known issue with the four node linear Cauchy finite elements. However, for the micropolar finite element that has been constructed here, the presence of an additional nodal degree of freedom associated with rotation and combined with a reduced integration scheme mitigates against this issue. Through appropriate discretization and checking numerical results against known scaling exponents and prefactors (see Table 2) for fracture toughness from the literature we account for this issue. The specimen dimensions correspond to a square region with size $L = 1$ mm containing a crack of length $2a$ representing 14% of the specimen width located in the middle ($\frac{2a}{L} = 0.14$), see Fig. 4. In order to simulate the mode I loading, we use mode I K-field asymptotic displacements given in (30) and (31). The FE model of the substitute micropolar continuum is subjected to vertical displacements at the top and the bottom edges. These vertical displacements generate a far field stress on the domain including a central crack shown in Fig. 4. Displacements on the left and right edges and rotations are not constrained. We use the asymptotic expressions of the displacement field given in (30) through (33) to find the values of the classical and micropolar stress intensity factors from the FE computations on the effective micropolar model. The critical value of K_{IC} is identified when the maximum local tensile stress (σ_{yy}) at a point slightly ahead of the crack tip located by $x/L = 0.0702, y/L = 0$ in Fig. 4, exceeds the failure stress σ_f of the parent solid material in tension. The choice of this location ahead of the crack tip is to ensure that the asymptotic expressions for the stress and displacement fields in (30) and (31) remain valid. Convergence of the computed maximum stress at the selected point ahead of crack tip is ensured by refining the mesh size. The higher order terms in the asymptotic series expansion, such as the T-stress (parallel to crack plane) of a Cauchy continuum, are not considered. In the classical Cauchy continuum T-stress has been shown to be significant (Fleck and Qiu, 2007) for strength considerations, particularly for a hexagonal lattice by switching the failure sites for uniaxial and biaxial loading conditions. We restrict the focus of this work is to show how proper accounting of the micropolar effects reconciles the scaling law for mode I toughness, K_{IC}. It will be useful to identify similar scaling relations for T-stress and higher order terms in the asymptotic expansions, in (30) and (31). Another feature worthy of a future study is anisotropy: the orientation dependent toughness of the crack relative to the lattice. Such studies are needed.

The fracture toughness in mode I, K_{IC}, is plotted against the relative density $\bar{\rho}$ in Fig. 5 for a hexagonal lattice where our micropolar continuum results are compared against literature. The relative density is restricted to $\bar{\rho} \leq 0.2$ in order to highlight the discrepancies among different predictions and available experimental data from the literature. Moreover the Timoshenko beam approximation itself is untenable at higher relative densities. It is clear that the exponent d equals 2 for hexagonal lattices consistent with beam theory calculations but it is much lower than other gradient theories of Chen et al. (1998). Comparing our results for a micropolar medium with those found in Fleck and Qiu (2007) within the classical elasticity framework, there is a very small difference between the pre-exponents D of the two

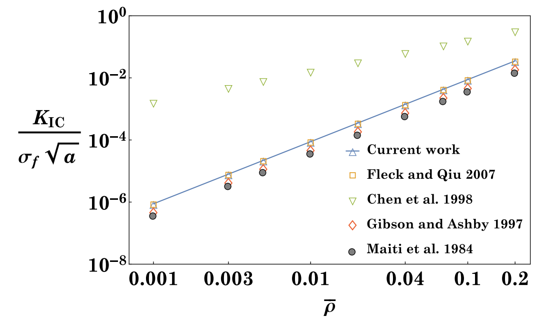

Fig. 5. Comparison of mode I fracture toughness calculated by micropolar FE with beam theory based scaling for a hexagonal lattice. At higher relative densities we expect non-affine bending deformation to be insignificant and the toughness assumes the value of the parent material of the lattice.

Table 2
Values of exponent d and pre-exponent D from relation (22) for different lattices in the current work, Fleck and Qiu (2007), Chen et al. (1998) and Cauchy continuum. A substitute micropolar continuum is used in the current work.

	Hexagonal		Triangular		Kagome	
	D	d	D	d	D	d
Current work	0.88 ± 0.01	2	0.45 ± 0.03	1	0.2 ± 0.01	0.5
Fleck and Qiu (2007)	0.80	2	0.50	1	0.21	0.5
Chen et al. (1998)	1.56	1	/	/	/	/
Cauchy	0.10	0	0.10	0	0.10	0

calculations. We further observe in Table 2 that the exponent d for our model is in good agreement with that of Fleck and Qiu (2007) and Maiti et al. (1984), and an earlier model by Chen et al.,Chen does not give a correct scaling. This validates the micropolar continuum finite element model. We again note that earlier gradient continuum formulations did not give correct scaling. This validation enables us to characterize the influence of topology on toughness within a micropolar continuum framework.

We used the micropolar FE model to calculate toughness based on Cauchy stress (K_{IC}) and couple-stress (L_{IC}) for the three lattice topologies over a range of relative densities $0.001 \leq \bar{\rho} \leq 0.2$ and the results are shown in Fig. 6. We already noted that the effective properties change with relative density and with topology in accordance with the relations summarized in the Table 1. It is clear from Fig. 6(a) that marked differences for toughness exist depending on the lattice topology at any given relative density. A perfect Kagome lattice exhibits higher K_{IC} and its elevated toughness is due to the transition from bending dominated deformation near the crack tip into a stretching dominated deformation far from the tip, shown in Fig. 7(c). The elastic boundary layers in lattices have been fully characterized in Phani and Fleck (2008) through a Bloch-wave formalism. By plotting the ratio of the bending to stretching strain energy versus the density in Fig. 7, we observe that the bending energy dominates below a threshold density $\bar{\rho} = 0.027$, whereas tension dominates above this density. One may accordingly consider this density to separate the bending from the stretching regime, from an energetic point of view.

Since the mode I fracture toughness of the elastic-brittle hexagonal lattice is expected to be dominated by the release of bending energy in the vicinity of the crack tip, one may expect that the micropolar stress intensity factor will play an important role in the formulation of the fracture energy criterion. Fracture in the bending dominated mode will occur when the maximum of the local bending moment reaches the bending strength m_f, itself depending upon the micropolar stress intensity factor L_{IC}; similar to the previous study of the scaling law of the classical stress intensity factor K_{1C} versus density, we will compute the scaling law of the couple stress intensity factor, according to the

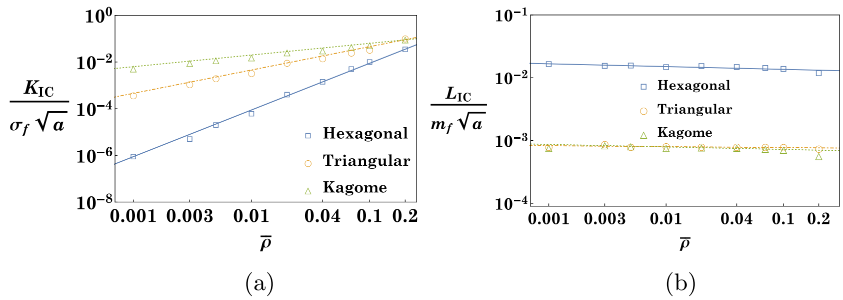

Fig. 6. The predicted mode I fracture toughness for (a) Cauchy stress K_{IC}; and (b) couple-stress L_{IC}. Note that L_{IC} is an order of magnitude higher for the bending dominated hexagonal lattice, while Kagome and triangular lattice have similar values for $0.001 \leq \bar{\rho} \leq 0.2$.

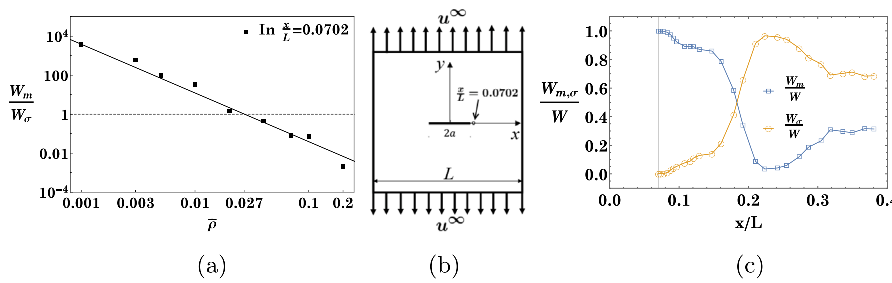

Fig. 7. (a) Ratio of bending to tensile energy $\frac{W_m}{W_\sigma}$ versus relative density $\bar{\rho}$ in micropolar equivalent continuum of a Kagome lattice at point $\frac{x}{L} = 0.0702$ near crack-tip, (b) Location of the point where the ratio of the strain energies is calculated as a function of the relative density, and (c) Spatial evolution from the crack tip of strain energies normalized by the total strain energy.

following relation:

$$\frac{L_{IC}}{m_f\sqrt{a}} = B\bar{\rho}^b \tag{34}$$

The couple stress intensity factor is of the same order as K_{IC} and decreases linearly with the density using double logarithmic axes as shown in Fig. 6(b); the scaling exponents b, and pre-exponents B, are identified from the FE simulations and relation (34) for the three lattices, the values of b and B are presented in Table 3. The hexagonal lattice has a nodal connectivity equal to 3 and is bending dominated which explains the lower values of both the classical and micropolar stress intensity factors for this lattice.

4.2. Strain energy release rate $\mathcal{G}$

The strain energy release rates can be obtained for each lattice topology by substituting their respective effective properties from Table 1 in the expression for the strain energy given in (33). We thus obtain the following expressions :

$$\begin{aligned}
\mathcal{G} &= \frac{4\pi}{E_s}\left[\frac{(4+3\bar{\rho}^2)(4+33\bar{\rho}^2)}{(4+9\bar{\rho}^2)^2}\cdot K_I^2 + \frac{a^2}{l^2}\frac{1}{\bar{\rho}^3}\cdot L_I^2\right], \quad \text{hexagonal lattice}\\
&= \frac{4\pi}{E_s}\left[\frac{24(84+5\bar{\rho}^2)}{\bar{\rho}(36+\bar{\rho}^2)^2}\cdot K_I^2 + 9\frac{a^2}{l^2}\frac{1}{\bar{\rho}^3}\cdot L_I^2\right], \quad \text{triangular lattice}\\
&= \frac{4\pi}{E_s}\left[\frac{12(36+5\bar{\rho}^2)}{\bar{\rho}(18+\bar{\rho}^2)^2}\cdot K_I^2 + \frac{9}{4}\frac{a^2}{l^2}\frac{1}{\bar{\rho}^3}\cdot L_I^2\right], \quad \text{kagome lattice.}
\end{aligned} \tag{35}$$

In order to quantify the significance of micropolar effects, we define a parameter ψ as a ratio of the two terms in (35) corresponding to the micropolar contribution (proportional to L_I^2) and the classical strain energy release rate (proportional to K_I^2) as:

$$\psi = \begin{cases}
\frac{(4+9\bar{\rho}^2)^2}{\bar{\rho}^3(4+3\bar{\rho}^2)(4+33\bar{\rho}^2)}\left(\frac{a}{l}\right)^2\left(\frac{L_I}{K_I}\right)^2, & \text{Hexagonal lattice}\\
\frac{9(36+\bar{\rho}^2)^2}{24\bar{\rho}^2(84+5\bar{\rho}^2)}\left(\frac{a}{l}\right)^2\left(\frac{L_I}{K_I}\right)^2, & \text{Triangular lattice}\\
\frac{9(18+\bar{\rho}^2)^2}{48\bar{\rho}^2(36+5\bar{\rho}^2)}\left(\frac{a}{l}\right)^2\left(\frac{L_I}{K_I}\right)^2, & \text{Kagome lattice}
\end{cases} \tag{36}$$

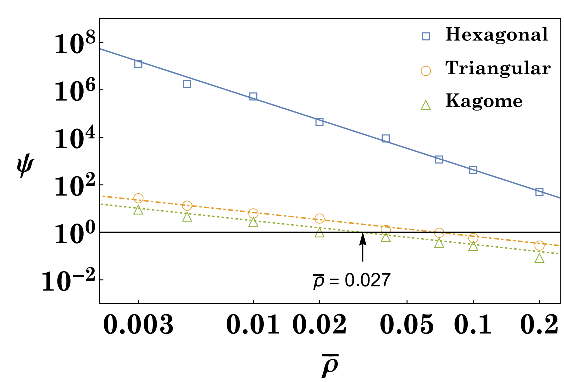

Fig. 8. Evolution of the classical to micropolar contributions to the energy release rate for the three lattices, relations (36), for $a/l = 7$. The horizontal solid line indicates the transition between bending to stretching dominated lattices in terms of strain energy release rate.

Table 3
Values of exponent b and pre-exponent B from relation (34) for different lattices in current work. The error bars are the standard deviations of the fitted straight line to the computational points.

	Hexagonal	Triangular	Kagome
B	0.01230 ± 0.00050	0.00070 ± 0.00002	0.00066 ± 0.000047
b	-0.04418 ± 0.00956	-0.01496 ± 0.00728	-0.04000 ± 0.015704

The bending part of the energy release rate corresponding to the micropolar behavior dominates in the non-affine regime (below the threshold value $\bar{\rho} = 0.027$) for the kagome lattice identified in Fig. 8 as the intersection of the scaling response of energy ratio with the horizontal transition line, whereas the classical contribution of the energy release rate dominates for densities above this threshold. The threshold for the rupture behavior to move to a regime dominated by the bending contribution of the energy release rate increases with the crack advancement: the bending energy contribution to rupture is enhanced when increasing the ratio of crack length to the beam length. Note that the crack length needs to be larger than beam length for continuum description to remain valid. For the hexagonal lattice which is bending dominated, the bending dominated regime transitions towards a tension dominated regime for higher densities in comparison to the two other lattices which are tension dominated. For the Kagome lattice, the density at which a transition towards a bending dominated rupture criterion occurs coincides with the density for which the bending to tensile energy ratio reaches unity, as shown in Fig. 7. We thus conclude that non-affine deformation and micropolar effects are orders of magnitude higher for a bending dominated hexagonal lattice. The micropolar framework for toughness developed here resolves the discrepancies between the gradient theory in Chen et al. (1998) and experimental (Gibson and Ashby, 1997; Maiti et al., 1984) and numerical observations (Fleck and Qiu, 2007). This finding supports the need to include the micropolar stress intensity factor into the energy release rate for the purposes of evaluating fracture toughness.

5. Concluding remarks

We applied the discrete asymptotic homogenization to elastic periodic lattices to obtain geometry governed effective elastic properties and used the resulting substitute micropolar continuum to study micropolar and topological effects on fracture toughness. The significance of moment intensities motivated us to reformulate the rupture criterion for both tensile (triangular, kagome) and bending dominated (hexagonal) lattice materials. The modified energy release rates incorporate a micropolar stress intensity factor into the rupture criterion of these lattices at the continuum level. This extends the classical rupture criterion based on the stress intensity factors to non-affine deformation regimes where the micropolar contribution to the strain energy release rate is shown to dominate. The scaling laws of the classical and micropolar mode I stress intensity factors with the network density have been computed. The following conclusions emerge from our study:

- The distinction between bending versus stretching dominated lattices extends to micropolar material constants as well.
- Bending dominated hexagonal lattices exhibit an order of magnitude higher micropolar material length scale compared to the stretching dominated kagome and triangular lattices.
- We obtain a micropolar elastic continuum that resolves the discrepancies between the gradient theory in Chen et al. (1998) and experimental (Gibson and Ashby, 1997; Maiti et al., 1984) and numerical observations (Fleck and Qiu, 2007).
- In the low relative density regime non-affine bending deformation field and couple-stress intensity cannot be ignored in bending dominated lattices.
- The ratio of energy release rates between the classical and the micropolar values is dependent on the relative density and lattice topology.

We restricted this study to isotropic lattices; anisotropic lattices as well as the mixed mode loading cases can be considered in a future study. Consideration of the effect of network plasticity on fracture toughness is also a topic for further study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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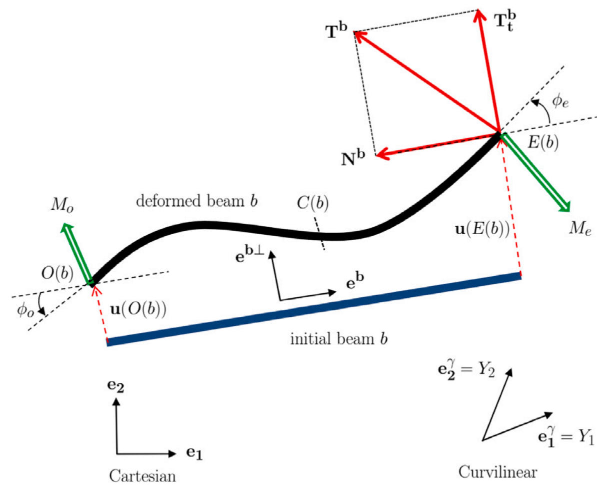

Fig. A.9. Geometric parameters characterizing the deformed beam.

Appendix A. A synopsis of discrete asymptotic homogenization

A.1. Homogenization of equilibrium equations

Our starting point is the set of equilibrium equations of a lattice whose parameters are presented in Fig. A.9. We use the assumption of small deformations as a consequence of the quasi-periodicity of the lattice after deformation.

The equilibrium equations of forces in virtual power formulation on the whole lattice can be written in the form of the following asymptotic expansions

$$\sum_{v^i \in \mathbb{Z}^3} \sum_{b \in B_R} \mathbf{T}^{\epsilon \mathbf{b}} \cdot [\mathbf{v}^\epsilon(O(b)) - \mathbf{v}^\epsilon(E(b))] = 0 \tag{A.1}$$

The equilibrium of the applied moments at nodes of the lattice writes as follows:

$$\sum_{v^i \in \mathbb{Z}^3} \sum_{b \in B_R} (\mathbf{M}^{O(b)} \cdot \mathbf{w}(O(b)) - \mathbf{M}^{E(b)} \cdot \mathbf{w}(E(b))) = 0 \tag{A.2}$$

We use this equation for the resolution of unknowns, but we can consider another approach that takes into account the balance of each beam in the lattice and which we can write in the form of asymptotic development in the center of each beam by

$$\sum_{v^i \in \mathbb{Z}^3} \sum_{b \in B_R} [\mathbf{M}^{O(b)\epsilon} \cdot \mathbf{w}^\epsilon(O(b)) - \mathbf{M}^{E(b)\epsilon} \cdot \mathbf{w}^\epsilon(E(b)) + \epsilon L^b (\mathbf{e}^b \times \mathbf{T}^{\epsilon b}) \cdot \mathbf{w}^\epsilon(C(b))] = 0. \tag{A.3}$$

The asymptotic development of the virtual velocity is given by

$$\mathbf{v}^\epsilon(O(\tilde{b})) - \mathbf{v}^\epsilon(E(\tilde{b})) = \mathbf{v}(\gamma^\epsilon) - \mathbf{v}(\gamma^\epsilon + \epsilon\delta^{ib}) = -\epsilon \frac{\partial \mathbf{v}(\gamma^\epsilon)}{\partial \gamma^i} \delta^{ib} + \dots \tag{A.4}$$

The virtual rotation rate field which is chosen sufficiently regular such that we have for the midpoint of each beam element

$$\mathbf{w}^{C(b)\epsilon} = \frac{1}{2}(\mathbf{w}^{E(b)\epsilon} + \mathbf{w}^{O(b)\epsilon}) \tag{A.5}$$

The rotation rate field is expected to be

$$\mathbf{w}^{O(b)\epsilon}(\gamma) = \mathbf{w}_0(\gamma) \tag{A.6}$$

This allows to obtain, after a Taylor first order development

$$\mathbf{w}^{E(b)\epsilon}(\gamma + \epsilon\delta^{ib}) \sim \mathbf{w}_0(\gamma) + \epsilon \frac{\partial \mathbf{w}_0(\gamma)}{\partial \gamma^i} \delta^{ib} \tag{A.7}$$

The final expressions of the equilibrium equations are given by

A.2. Equilibrium of forces

$$\begin{aligned}&\sum_{v^i\in\mathbb{Z}^3}\sum_{b\in B_R}\Big[\epsilon^2\Big(E_s\eta(\mathbf{e}^b\cdot\Delta\mathbf{U}_1)\mathbf{e}^b+\Big(E_s\eta^3\mathbf{e}^{b\perp}\cdot\Delta\mathbf{U}_1\\&-\tfrac12E_s\eta^3L^b\Big(\phi_0^{O_R(b)}+\phi_0^{E_R(b)}\Big)\Big)\mathbf{e}^{b\perp}\Big)\cdot\tfrac{\partial\mathbf{v}(\gamma^\epsilon)}{\partial\gamma^i}\delta^{ib}\\&+\epsilon^3\Big(E_s\eta(\mathbf{e}^b\cdot\Delta\mathbf{U}_2)\mathbf{e}^b+\Big(E_s\eta^3\mathbf{e}^{b\perp}\cdot\Delta\mathbf{U}_2-\tfrac12E_s\eta^3L^b\cdot\\&\Big(\phi_1^{O_R(b)}+\phi_1^{E_R(b)}+\tfrac{\partial\phi_0}{\partial\gamma^i}\delta^{ib}\Big)\Big)\mathbf{e}^{b\perp}\Big)\cdot\tfrac{\partial\mathbf{v}(\gamma^\epsilon)}{\partial\gamma^i}\delta^{ib}\Big]=0\end{aligned}\tag{A.8}$$

We then use the same result following Mourad (2003) to transform the discrete formulation of the equilibrium equation into a continuous formulation. The equation (A.8) becomes after homogenization

$$\int_\Omega\mathbf{S}^{\mathbf{i}}\cdot\frac{\partial\mathbf{v}(\gamma)}{\partial\gamma^i}d\gamma=0\quad\text{and}\quad\mathbf{S}^{\mathbf{i}}\sim\mathbf{S}_1^i+\epsilon\mathbf{S}_{\mathbf{2}}^{\boldsymbol{i}}\tag{A.9}$$

Stress vectors $\mathbf{S}_1^i$ and $\mathbf{S}_2^i$ are drawn from the equation (A.8) such that

$$\begin{aligned}\mathbf{S}_1^i&=\textstyle\sum_{b\in B_R}\epsilon^2\Big(E_s\eta(\mathbf{e}^b\cdot\Delta\mathbf{U}_1)\mathbf{e}^b+\Big(E_s\eta^3\mathbf{e}^{b\perp}\cdot\Delta\mathbf{U}_1\\&\quad-\tfrac12E_s\eta^3L^b\Big(\phi_0^{O_R(b)}+\phi_0^{E_R(b)}\Big)\Big)\mathbf{e}^{b\perp}\Big)\\\mathbf{S}_2^i&=\textstyle\sum_{b\in B_R}\epsilon^3\Big(E_s\eta(\mathbf{e}^b\cdot\Delta\mathbf{U}_2)\mathbf{e}^b+\Big(E_s\eta^3\mathbf{e}^{b\perp}\cdot\Delta\mathbf{U}_2\\&\quad-\tfrac12E_s\eta^3L^b\Big(\phi_1^{O_R(b)}+\phi_1^{E_R(b)}+\tfrac{\partial\phi_0}{\partial\gamma^i}\delta^{ib}\Big)\Big)\mathbf{e}^{b\perp}\Big)\end{aligned}\tag{A.10}$$

A.3. Equilibrium of moments

The equilibrium equation of moments in virtual power format is given by (A.2). The asymptotic development of the variables involved in this equation leads to the following equilibrium equation :

$$\begin{aligned}&\textstyle\sum_{v^i\in\mathbb{Z}^3}\sum_{b\in B_R}\epsilon^2\Big[\Big(E_s\eta^3\epsilon^2\frac{(L^b)^2}{6}(2\phi^{O(b)\epsilon}+\phi^{E(b)\epsilon})\\&-\tfrac12E_s\eta^3\epsilon L^b\mathbf{e}^{b\perp}\cdot\Delta\mathbf{U}_1\Big)\cdot w_0+\Big(E_s\eta^3\epsilon^2\tfrac{(L^b)^2}{6}(\phi^{O(b)\epsilon}+\\&2\phi^{E(b)\epsilon})-\tfrac12E_s\eta^3\epsilon L^b\mathbf{e}^{b\perp}\cdot\Delta\mathbf{U}_1\Big)\cdot w_0+L^{b0}\Big(E_s\eta^3\epsilon\mathbf{e}^{b\perp}\cdot\\&\Delta\mathbf{U}_1-\tfrac12E_s\eta^3\epsilon^2L^b(\phi^{O(b)\epsilon}+\phi^{E(b)\epsilon})\Big)\cdot w_0\Big]+\\&\epsilon^3\Big[\Big(E_s\eta^3\epsilon^2\tfrac{(L^b)^2}{6}(2\phi^{O(b)\epsilon}+\tfrac{\partial\phi_0}{\partial\gamma^i}\delta^{ib}+\phi^{E(b)\epsilon})-\\&\tfrac12E_s\eta^3\epsilon L^b\mathbf{e}^{b\perp}\cdot\Delta\mathbf{U}_2\Big)\cdot w_0+\Big(E_s\eta^3\epsilon^2\tfrac{(L^b)^2}{6}(\phi^{O(b)\epsilon}+\\&2\tfrac{\partial\phi_0}{\partial\gamma^i}\delta^{ib}+2\phi^{E(b)\epsilon})-\tfrac12E_s\eta^3\epsilon L^b\mathbf{e}^{b\perp}\cdot\Delta\mathbf{U}_2\Big)\cdot w_0\\&+L^{b0}\Big(E_s\eta^3\epsilon\mathbf{e}^{b\perp}\cdot\Delta\mathbf{U}_2-\tfrac12E_s\eta^3\epsilon^2L^b(\phi^{O(b)\epsilon}+\tfrac{\partial\phi_0}{\partial\gamma^i}\delta^{ib}+\\&\phi^{E(b)\epsilon})\Big)\cdot w_0+\Big(E_s\eta^3\epsilon^2\tfrac{(L^b)^2}{6}(\phi^{O(b)\epsilon}+2\phi^{E(b)\epsilon})-\\&\tfrac12E_s\eta^3\epsilon L^b\mathbf{e}^{b\perp}\cdot\Delta\mathbf{U}_1\Big)\cdot\tfrac{\partial w_0}{\partial\gamma^i}\delta^{ib}+\tfrac{L^{b0}}{2}\Big(E_s\eta^3\epsilon\mathbf{e}^{b\perp}\cdot\Delta\mathbf{U}_1-\\&\tfrac12E_s\eta^3\epsilon^2L^b(\phi^{O(b)\epsilon}+\phi^{E(b)\epsilon})\Big)\cdot\tfrac{\partial w_0}{\partial\gamma^i}\delta^{ib}\Big]+\\&\epsilon^4\Big[\Big(E_s\eta^3\epsilon^2\tfrac{(L^b)^2}{6}(\phi^{O(b)\epsilon}+2\tfrac{\partial\phi_0}{\partial\gamma^i}\delta^{ib}+2\phi^{E(b)\epsilon})-\\&\tfrac12E_s\eta^3\epsilon L^b\mathbf{e}^{b\perp}\cdot\Delta\mathbf{U}_2\Big)\cdot\tfrac{\partial w_0}{\partial\gamma^i}\delta^{ib}+\tfrac{L^{b0}}{2}\Big(E_s\eta^3\epsilon\mathbf{e}^{b\perp}\cdot\Delta\mathbf{U}_2-\\&\tfrac12E_s\eta^3\epsilon^2L^b(\phi^{O(b)\epsilon}+\tfrac{\partial\phi_0}{\partial\gamma^i}\delta^{ib}+\phi^{E(b)\epsilon})\Big)\cdot\tfrac{\partial w_0}{\partial\gamma^i}\delta^{ib}\Big]=0\end{aligned}\tag{A.11}$$

Terms with the higher powers in ϵ^3 and ϵ^4 are ignored. To move to the continuous formulation, we use the same technique as for the forces, and by letting $\epsilon\to0$, equation (A.11) can be written in the form

$$\int_\Omega\boldsymbol{\mu}^{\boldsymbol{i}}\cdot\frac{\partial\mathbf{w}(\gamma)}{\partial\gamma^i}d\gamma=0\quad\text{and}\quad\boldsymbol{\mu}^{\boldsymbol{i}}\sim\epsilon\boldsymbol{\mu}_1^i+\epsilon^2\boldsymbol{\mu}_{\mathbf{2}}^{\boldsymbol{i}}\tag{A.12}$$

The couple stress vectors are derived from the equation (A.11) and we write

$$\boldsymbol{\mu}_1^i=\sum_{b\in B_R}\epsilon\Big(E_s\eta^3\frac{(L^b)^2}{12}\big(\phi_0^{E_R(b)}-\phi_0^{O_R(b)}\big)\Big)\delta^{ib}\tag{A.13}$$

$$\boldsymbol{\mu}_2^i=\sum_{b\in B_R}\epsilon^2\Big(E_s\eta^3\frac{(L^b)^2}{12}\big(\phi_0^{E_R(b)}-\phi_0^{O_R(b)}+\frac{\partial\phi_0}{\partial\gamma^i}\delta^{ib}\big)\Big)\delta^{ib}\tag{A.14}$$

The equilibrium equations of forces and moments will be solved on two separate orders. The strain tensor is responsible for the variations of the variables at the first order $(\mathbf{u}_1^n,\phi_0^n)$, while the micro-curvature tensor is responsible for the existence of the second-order variables $(\mathbf{u}_2^n,\phi_1^n)$. The same form as the equations of the linear elasticity of a micropolar medium (Cosserat) is obtained

$$\boldsymbol{\sigma}\sim\frac1g\mathbf{S}_1^i\otimes\frac{\partial\mathbf{R}}{\partial\gamma^i}+\frac1g\epsilon\mathbf{S}_2^i\otimes\frac{\partial\mathbf{R}}{\partial\gamma^i}\tag{A.15}$$

$$\mathbf{m}\sim\frac1g\epsilon\boldsymbol{\mu}_1^i\otimes\frac{\partial\mathbf{R}}{\partial\gamma^i}+\frac1g\epsilon^2\boldsymbol{\mu}_2^i\otimes\frac{\partial\mathbf{R}}{\partial\gamma^i}\tag{A.16}$$

Appendix B. Stiffness matrices

The effective moduli are evaluated as closed form expressions for the hexagonal lattice of beam length l and relative density $\bar\rho$,

$$\frac{E_s\bar\rho}{4}\begin{bmatrix}\frac{4+9\bar\rho^2}{4+3\bar\rho^2}&\frac{4-3\bar\rho^2}{4+3\bar\rho^2}&0&0&0&0\\\frac{4-3\bar\rho^2}{4+3\bar\rho^2}&\frac{4+9\bar\rho^2}{4+3\bar\rho^2}&0&0&0&0\\0&0&\frac34\bar\rho\left(\frac{12+3\bar\rho^2}{4+3\bar\rho^2}\right)&\frac34\bar\rho\left(\frac{4-3\bar\rho^2}{4+3\bar\rho^2}\right)&0&0\\0&0&\frac34\bar\rho\left(\frac{4-3\bar\rho^2}{4+3\bar\rho^2}\right)&\frac34\bar\rho\left(\frac{12+3\bar\rho^2}{4+3\bar\rho^2}\right)&0&0\\0&0&0&0&\frac{l^2}{8}\bar\rho^2&0\\0&0&0&0&0&\frac{l^2}{8}\bar\rho^2\end{bmatrix}\tag{B.1}$$

for the triangular lattice

$$\frac{E_s\bar\rho}{8}\begin{bmatrix}3+\frac{\bar\rho^2}{12}&1-\frac{\bar\rho^2}{12}&0&0&0&0\\1-\frac{\bar\rho^2}{12}&3+\frac{\bar\rho^2}{12}&0&0&0&0\\0&0&1+\frac{\bar\rho^2}{4}&1-\frac{\bar\rho^2}{12}&0&0\\0&0&1-\frac{\bar\rho^2}{12}&1+\frac{\bar\rho^2}{4}&0&0\\0&0&0&0&\frac{l^2}{36}\bar\rho^2&0\\0&0&0&0&0&\frac{l^2}{36}\bar\rho^2\end{bmatrix}\tag{B.2}$$

and for the kagome lattice

$$\frac{E_s\bar\rho}{16}\begin{bmatrix}6+\frac{\bar\rho^2}{3}&2-\frac{\bar\rho^2}{3}&0&0&0&0\\2-\frac{\bar\rho^2}{3}&6+\frac{\bar\rho^2}{3}&0&0&0&0\\0&0&2+5\frac{\bar\rho^2}{3}&2-\bar\rho^2&0&0\\0&0&2-\bar\rho^2&2+5\frac{\bar\rho^2}{3}&0&0\\0&0&0&0&\frac{2l^2}{9}\bar\rho^2&0\\0&0&0&0&0&\frac{2l^2}{9}\bar\rho^2\end{bmatrix}\tag{B.3}$$

Appendix C. Validation of the micropolar model

We motivate the micropolar representation of lattices by determining a closed-form solution of the microrotation for a tensile test on a square specimen of the hexagonal lattice in Fig. C.10. We compare the local microrotations inside the square specimen with the fiber rotations predicted in the same domain by a fully resolved Finite Element (FE) computation. To derive the closed-form solution, the material within the domain is modeled by a homogeneous micropolar continuum whose effective properties are computed by discrete homogenization elucidated above. As to the adopted boundary conditions (Fig. C.10), the microrotations are assumed to be zero everywhere, except on the traction side (on the right) where we impose a displacement u_0 and a uniform microrotation ϕ_0 in order to simulate a simple uniaxial test of

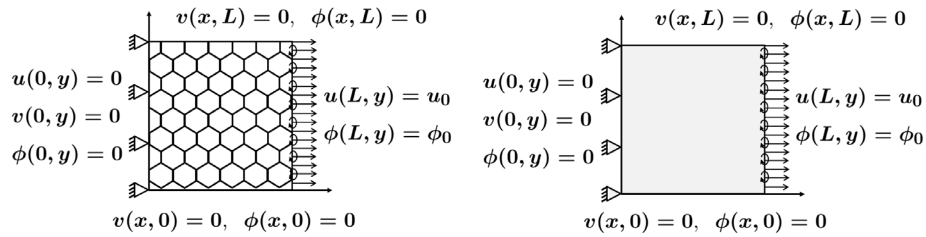

Fig. C.10. Hexagonal lattice and an equivalent micropolar behavior subjected to the same boundary conditions for a simple traction test.

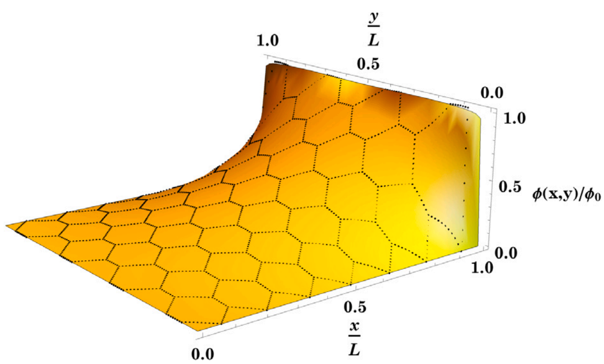

Fig. C.11. Comparison between the closed form solution (continuous surface) and microstructural finite element computation based solutions (black points) of the normalized microrotations field $\phi(x,y)/\phi_0$ for the tensile loading in Fig. C.10.

a micropolar continuum. We recall that the strain and microcurvature tensors in the framework of 2D micropolar elasticity are given by

$$e=\begin{bmatrix}\frac{\partial v}{\partial x}-\phi & \frac{\partial v}{\partial x}-\phi & 0\\ \frac{\partial u}{\partial y}+\phi & \frac{\partial v}{\partial y} & 0\\ 0 & 0 & 0\end{bmatrix} \text{ and } \kappa=\begin{bmatrix}0&0&0\\0&0&0\\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & 0\end{bmatrix} \tag{C.1}$$

The insertion of kinematic variables Eqs. (1) and (2) in the micropolar constitutive laws presented in Eqs. (7) and (8), leads to the following expression of the stress and couple stress components

$$\begin{aligned}
\sigma_{xx} &= (\lambda+2\mu)\frac{\partial u}{\partial x}+\lambda\frac{\partial v}{\partial y}\\
\sigma_{yy} &= \lambda\frac{\partial u}{\partial x}+(\lambda+2\mu)\frac{\partial v}{\partial y}\\
\sigma_{xy} &= 2\mu\left(\frac{\partial v}{\partial x}-\phi\right)\\
\sigma_{yx} &= 2\mu\left(\frac{\partial u}{\partial y}+\phi\right)\\
m_{xz} &= 2\beta\frac{\partial \phi}{\partial x}\\
m_{yz} &= 2\beta\frac{\partial \phi}{\partial y}
\end{aligned} \tag{C.2}$$

We then insert these resulting expressions into the equilibrium Eqs. (3) and (4), leading to the following harmonic equation for the microrotation variable. This differential equation does not involve the computed homogenized micropolar coefficients as a consequence of isotropy.

$$\frac{\partial^2 \phi}{\partial x^2}+\frac{\partial^2 \phi}{\partial y^2}=0 \tag{C.3}$$

We assume that the solution of this 2D Laplace equation can be written in a separable form

$$\phi(x,y)=f(x)\cdot g(y) \tag{C.4}$$

Using the Fourier transform, we end up with the analytical solution of the Laplace equation for the microrotation field

$$\frac{\phi}{\phi_0}=\frac{4}{\pi}\sum_{m=1}^{\infty}\frac{1}{(2m-1)\sinh[(2m-1)\pi]}\sinh\left[\frac{(2m-1)\pi x}{L}\right]\sin\left[\frac{(2m-1)\pi y}{L}\right] \tag{C.5}$$

This validation has also been verified by Berkache et al. (2019) for random fiber networks. The microrotation field $\phi(x,y)$ predicted by the analytical solution of the micropolar equivalent continuum is in good agreement with that of the fully resolved finite element microstructural computations over the hexagonal lattice (Fig. C.11). This provides a justification of the employed micropolar substitution continuum.

Appendix D. List of symbols

$\boldsymbol{A_e}$	Cauchy stress stiffness matrix
$\boldsymbol{D_e}$	Couple stress stiffness matrix
δ^i	Shift factor for nodes belonging to a neighboring cell
E_s	Elastic modulus of structural material
E^*	Homogenized elastic modulus
$\boldsymbol{e}_i^{\gamma}$	Unit vectors of the curvilinear coordinate system associated to the lattice
$\boldsymbol{e_i}$	Unit vectors of the cartesian coordinate system
$\boldsymbol{e}$	Deformation tensor
ϵ	Small scale parameter
ε_{ijk}	Levi-Civita permutation tensor
γ	Micropolar constant
γ^i	curvilinear coordinates associated with the vectors $\boldsymbol{e}_i^{\gamma}$
g	Jacobian of the transformation from dx to $d\gamma$
$\mathcal{G}$	Strain energy release rate
I	Moment of inertia of the fiber cross-section
$\boldsymbol{\kappa}$	Curvature tensor
k_l	Axial stiffness of a beam
k_f	Bending stiffness of a beam
L	Characteristic length of the whole structure
l	Length of the beam
$\boldsymbol{m}$	Couple stress tensor
M^n	Moment at node n
μ	Classical Lamé shear modulus
μ_c	Micropolar constant
$\boldsymbol{\mu}_i^j$	Couple stress vectors
$\boldsymbol{N}^b$	Normal forces of the beam b
ν^*	Homogenized Poisson's ratio
ϕ^n	Angular displacement at node
ϕ_0	Homogenized angular displacement of the equivalent medium at zeroth order
$\boldsymbol{\sigma}$	Cauchy stress tensor
O,E	Origin and end node of a beam respectively
$\boldsymbol{R}$	Continuum node position
$\boldsymbol{S}^i$	Stress vectors
t	Width of the beam
$\boldsymbol{T}^b$	Resultant of forces at the nodes of a beam b
$\boldsymbol{T}_t^b$	Shear forces of the beam b
$\boldsymbol{U}^n(u^n,v^n)$	Displacement vector at node n
w	Thickness of the beam
$\boldsymbol{x}^i$	Coordinates associated with vectors $\boldsymbol{e_i}$

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